i.e. we compose the flows of

$$\dot{\boldsymbol{\pi}} = \boldsymbol{\pi} \times \begin{bmatrix} T_1^{-1}\pi_1 \\ 0 \\ 0 \end{bmatrix}, \qquad \dot{\boldsymbol{\pi}} = \boldsymbol{\pi} \times \begin{bmatrix} 0 \\ T_2^{-1}\pi \\ 0 \end{bmatrix}, \qquad \dot{\boldsymbol{\pi}} = \boldsymbol{\pi} \times \begin{bmatrix} 0 \\ 0 \\ T_3^{-1}\pi \end{bmatrix},$$

Each of these is, it turns out, easily integrated as a planar rotation (see Exercise 7). This yields a simple, efficient algorithm for the free rigid body which is symplectic in a generalized sense.

4.4.2 Rigid Body Molecular Dynamics Algorithm

To develop an integrator for the complete system (including external forces), we invoke the splitting technique,[2] solving successively for the translational motion from the center of mass equations, integrating the torques due to the rotational interactions, then solving the Euler equations, and recovering the rotation matrix from the linear system

$$\dot{\boldsymbol{\Theta}} = \boldsymbol{\Theta}\, \mathrm{skew}(\boldsymbol{T}^{-1}\boldsymbol{\pi}).$$

This is the approach described in [112] and is implemented in several software packages.

The reliability of these methods is supported by the fact that the combined scheme is a symplectic method in the generalized sense of constrained dynamics.

The specific steps of the algorithm proposed in [112] are given here. In what follows, we denote the variables associated to the kth rigid body by $\boldsymbol{q}^k$ (position of center of mass), $\boldsymbol{\Theta}^k$ (3×3 rotation matrix), $\boldsymbol{p}^k$ (linear momentum vector), and $\boldsymbol{\pi}^k$ (angular momentum vector). Individual components are defined by double subscripts: the 2nd component of $\boldsymbol{\pi}^k$ is π_2^k. Procedures given below for rigid body k are meant to be performed for all rigid bodies in the system.

The algorithm is initialized by computing the forces $\boldsymbol{F}^k$ and torques $\boldsymbol{\tau}^k$ acting on the kth rigid body, evaluated at the initial point. The relevant expressions are:

$$\boldsymbol{F}^k = -\frac{\partial U}{\partial \boldsymbol{q}^k}, \qquad \boldsymbol{\tau}^k = -\mathrm{rot}\left(\left[\boldsymbol{\Theta}^k\right]^T \frac{\partial U}{\partial \boldsymbol{\Theta}^k} \right).$$

[2] We note the seminal role of the work in astronomy of Touma and Wisdom [368] in elucidating the "divide and conquer" approach for deriving such schemes for rigid bodies.